Experiment 05 — Spectral Lag and Latent-Proxy Corrections

Two-Layer CTMC experimental framework

FINAL ANALYSIS — PRESPECIFIED 100-REPLICATION SAMPLE

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Abstract

This experiment tests two candidate aggregate-descriptor corrections to premature inter-community leakage in MicroMacro simulation. A spectral lag delays source contagiousness, while a latent-boundary proxy relaxes toward aggregate prevalence. Full Micro is the reference and all five variants share the network and initial condition within a replication. Inference uses the replication as the independent unit and distinguishes completed-only Wasserstein metrics from censoring-aware all-run RMST metrics. The present document reports the final stage. The analysis uses the prespecified replication IDs 1–100 for every scenario and variant.

1 Research question

Can spectral lag, latent-proxy relaxation, or their combination reduce sparse chain-of-cliques terminal-arrival bias without knowing bridge-node identities?

2 Motivation

Section 3.7 of *Notes on Micro–Macro Epidemic Networks* reports premature leakage in the sparse benchmark. Experiment 01 separated hazard and placement effects but left bridge-specific error. The corrections tested here are *candidate corrections proposed within the experimental framework, motivated by spectral epidemic-threshold bounds*; they are not attributed to the article.

3 Experimental design

The five paired variants are V0 Full Micro, V1 production aggregate MicroMacro, V2 lag-shifted MicroMacro, V3 latent-proxy MicroMacro, and V4 combined. The baseline has $L = 10$, $n = 100$, $k = 5$, mean internal degree approximately 6, $\beta_{in} = \beta_{out} = 0.003$, $\gamma = 0$, $\tau_{micro} = 1$, and $T_{end} = 4000$. Mini-sweeps vary $k \in \{1, 5, 10\}$ and $d \in \{4, 6, 10\}$ without taking a Cartesian product. Network and initialization streams are shared within replication; event streams are variant-specific. This is a matched block design, not pathwise common-random-number pairing.

4 Mathematical definitions

For community i, N_i is its size, $i_i(t)$ its infected count, $\bar{i}_i(t) = i_i(t)/N_i$, and $\bar{s}_j(t)$ is normalized susceptibility. The reference macro hazard is

$$\lambda_{ij}^{ref}(t) = \beta_{out} W_{ij} \bar{i}_i(t) \bar{s}_j(t).$$

The lag is

$$\tau_{find}(i) = \frac{\ln N_i}{\beta_{in}(\lambda_1(G_i) - 1)}, \qquad \bar{i}_i^{lag}(t) = \bar{i}_i(\max(0, t - \tau_{find}(i))),$$

and $\lambda_{ij}^A = \beta_{out} W_{ij} \bar{i}_i^{lag} \bar{s}_j$. The latent rate and analytical synchronization update are

$$\kappa(i) = \frac{\beta_{in}(\lambda_1(G_i) - 1)}{\ell(G_i)}, \qquad \bar{i}_i^B(t + \tau) = \bar{i}_i^B(t) + (1 - e^{-\kappa(i)\tau})(\bar{i}_i(t) - \bar{i}_i^B(t)),$$

with $\bar{i}_i^B(0) = 0$ and $\lambda_{ij}^B = \beta_{out} W_{ij} \bar{i}_i^B \bar{s}_j$. V4 applies the same latent update to $\bar{i}_i^{lag}$. If $\lambda_1 \leq 1$ or ℓ is undefined, corrected outgoing hazard is zero and the descriptors are flagged degenerate. Arrival definitions are $T_i = \inf\{t : i_i(t) > 0\}$, $T_{last} = T_{L-1}$, and $\Delta T_i = T_{i+1} - T_i$; censored values equal T_{end} with a separate observation flag.

5 Parameters

Each configured scenario uses a connected fixed-edge-count Erdős–Rényi community, uniform susceptible import placement, and one uniformly selected seed in community zero. Graph descriptors are computed once before simulation.

6 Implementation details

Adjacency spectral radius is computed by shifted power iteration to tolerance 10^{-6} with at most 1000 iterations and a residual-norm convergence check. Mean shortest-path length is computed on the unweighted local graph. Lag history uses fixed synchronization points and linear interpolation. The latent ODE uses the analytical exponential update, not Euler integration. No descriptor fallback is permitted. Outputs retain deterministic stream metadata and per-replication graph checksums. Integrated hazard suppression measures how strongly each correction changes V1 exposure, while synchronized correlation between $\bar{i}_i$ and the latent state checks relaxation semantics. The proxy uses no boundary-node identities and must not be interpreted as a direct bridge-aware state.

7 Sanity checks

Table 1: Recorded stage gates and actual diagnostic values.

Gate	Value	Status
Historical reference	20 archived replays; max arrival difference 4.55e-13	passed
Formula and identity checks	descriptor ordering, degenerate case, scheduler identity, and zero-lag limit	passed
Final-sample integrity	25 scenario-variant cells; replication IDs 1–100 present in every cell	passed
Schema and reproducibility	required columns present; recorded deterministic payload checks passed	passed
Production-code audit	before/after hashes identical; four pre-existing uncommitted file diffs	warning

Interpretation below is conditional on the formula, reproducibility, pairing, reference, schema, and production-immutability gates recorded in the manifest. The production-immutability flag reflects pre-existing uncommitted production-code changes: the recorded before/after hashes within this run are identical. It is an audit warning, not evidence that the simulation changed its own code while running.

8 Results tables

Table 1 reports the primary all-run restricted W_1 and RMST metrics. The completed-only relative W_1 is secondary and is shown only beside its completion denominator. Baseline raw data also contain residual IDs above 100 from an older interrupted launch; this report excludes them and uses IDs 1–100 uniformly.

Scenario	Variant	Completed	restricted W_1 [95% CI]	completed rel. W_1	ΔRMST vs V0 [95% CI]	ΔRMST vs V1
baseline	Full Micro	99/100	0 [0, 0]	0	0 [0, 0]	-200.6
baseline	Baseline MicroMacro	98/100	208.2 [112.9, 319]	0.06939	200.6 [98.16, 314.6]	0
baseline	Lag	0/100	1071 [995.3, 1146]	NA	1071 [993, 1151]	870.3
baseline	Latent proxy	44/100	905.8 [816.9, 998.7]	0.2421	905.8 [813.7, 992.6]	705.2
baseline	Lag + latent	0/100	1071 [994.9, 1146]	NA	1071 [995.3, 1146]	870.3
d10	Full Micro	100/100	0 [0, 0]	0	0 [0, 0]	-165.1
d10	Baseline MicroMacro	100/100	169.1 [119.8, 240.4]	0.0848	165.1 [88.49, 240.7]	0
d10	Lag	96/100	1513 [1428, 1592]	0.7484	1513 [1433, 1592]	1347
d10	Latent proxy	100/100	718.7 [625.3, 812.2]	0.3605	718.7 [628.5, 808.4]	553.6
d10	Lag + latent	58/100	1874 [1804, 1936]	0.8916	1874 [1807, 1937]	1709
d4	Full Micro	42/100	0 [0, 0]	0	0 [0, 0]	-34.45
d4	Baseline MicroMacro	35/100	37.85 [24.95, 126.4]	0.02288	34.45 [-53.37, 125.8]	0
d4	Lag	0/100	184.4 [125.1, 252.6]	NA	184.4 [124.9, 250.9]	150
d4	Latent proxy	0/100	184.4 [128.5, 249.5]	NA	184.4 [127.9, 250.3]	150
d4	Lag + latent	0/100	184.4 [123.1, 254.9]	NA	184.4 [124.8, 254.2]	150
k1	Full Micro	1/100	0 [0, 0]	0	0 [0, 0]	2.594
k1	Baseline MicroMacro	1/100	2.594 [0, 17.25]	0.0704	-2.594 [-17.25, 9.471]	0
k1	Lag	0/100	3.157 [0, 9.471]	NA	3.157 [0, 9.471]	5.751
k1	Latent proxy	0/100	3.157 [0, 9.471]	NA	3.157 [0, 9.471]	5.751
k1	Lag + latent	0/100	3.157 [0, 9.471]	NA	3.157 [0, 9.471]	5.751
k10	Full Micro	100/100	0 [0, 0]	0	0 [0, 0]	-113.4
k10	Baseline MicroMacro	100/100	114.6 [58.64, 192.6]	0.0519	113.4 [34.06, 194.2]	0
k10	Lag	2/100	1791 [1727, 1854]	0.7792	1791 [1731, 1851]	1678
k10	Latent proxy	100/100	1029 [939.6, 1119]	0.4662	1029 [938.4, 1114]	915.7
k10	Lag + latent	0/100	1792 [1726, 1855]	NA	1792 [1730, 1855]	1679

9 Figures

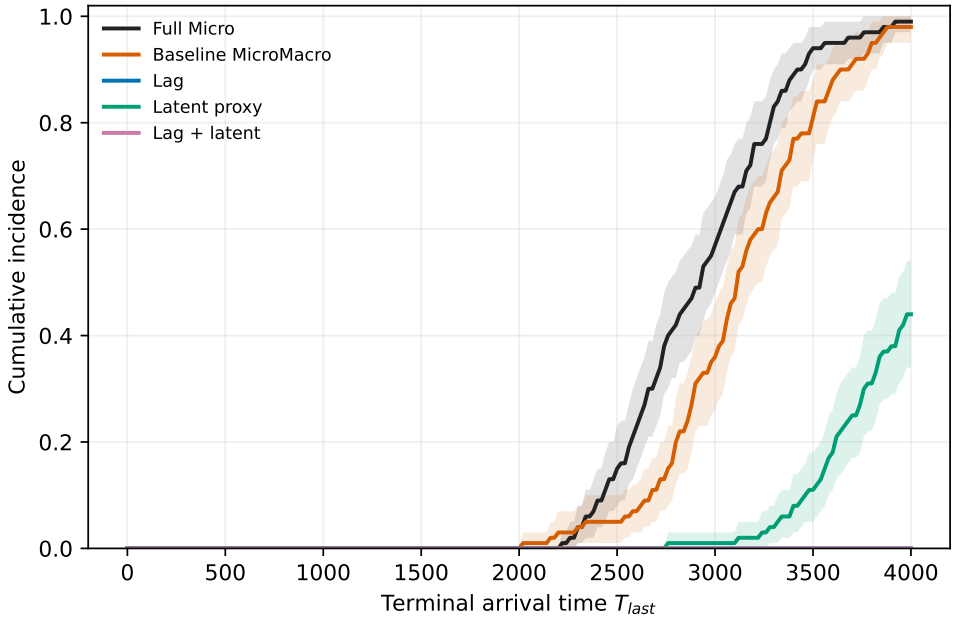


Figure 1: Baseline cumulative incidence of terminal arrival. Curves retain all runs and terminate at the observed completion probability; bands are pointwise 95% replication-bootstrap intervals.

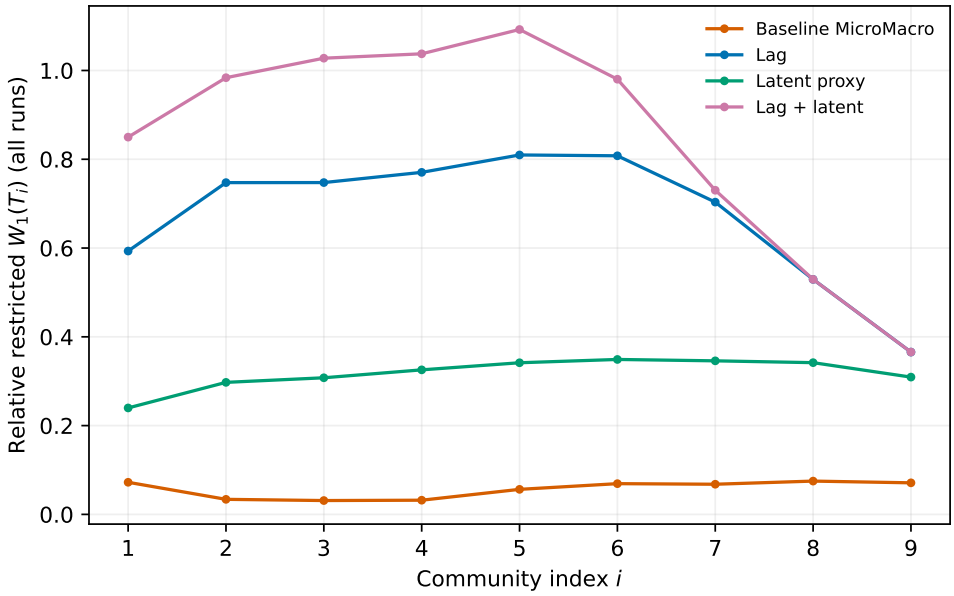


Figure 2: Layer-wise all-run restricted relative $W_1(T_i)$ against V0. Censored arrivals retain the common administrative horizon.

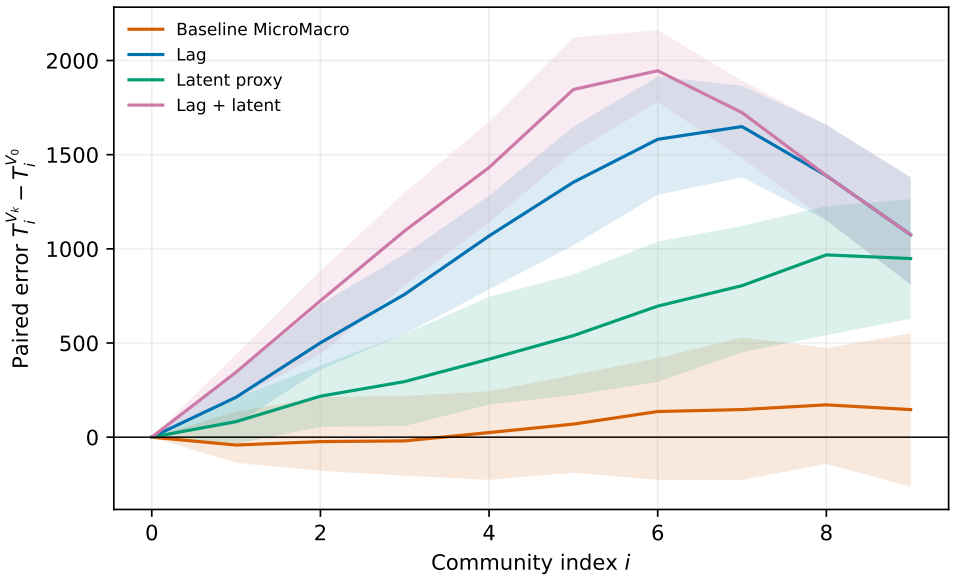


Figure 3: Paired cumulative arrival error (variant minus V0), median and IQR over replications; censored arrivals retain the common horizon.

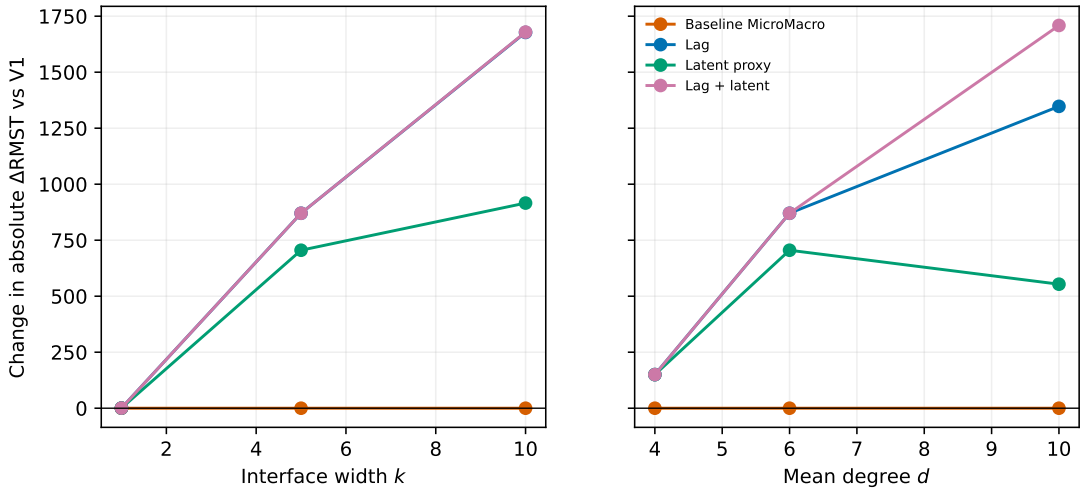


Figure 4: Change in absolute restricted timing discrepancy relative to V1 across the interface-width and density mini-sweeps. Positive values mean that a correction worsened the V1 discrepancy. Baseline supplies the shared $k = 5, d = 6$ point.

10 Uncertainty analysis

All confidence intervals use 2000 paired bootstrap resamples of replication IDs. Event rows are never bootstrap units. Primary W_1 compares the all-run horizon-restricted terminal-arrival distributions; completed-only W_1 is a secondary conditional diagnostic normalized by the V0 completed mean. ΔRMST is variant minus V0 using all horizon-restricted runs. Equivalence requires the entire ΔRMST interval to lie within $\pm 0.05\,\mathrm{mean}(T_{last}^{V0})$.

11 Scientific interpretation

H1 is assessed by V2 versus V1, H2 by V3 versus V1, and their joint behavior by V4. The replication-level factorial interaction $V4-V2-V3+V1$ on restricted terminal arrival is -705.2 (95% CI -798.1 to -608.8). Claims of improvement require both passed gates and the reported uncertainty intervals; an interval containing zero alone is not evidence of equivalence. Because several terminal cells saturate at the administrative horizon, the terminal interaction is descriptive rather than a clean additivity test. In the prespecified baseline sample, V1 completed in 98% of runs and was later than V0 by ΔRMST = 200.6 (95% CI 98.16 to 314.6). V2 and V4 had zero terminal completions and all-run ΔRMST values 1071 and 1071. V3 completed in 44% and had ΔRMST = 905.8. All literal corrections therefore increased delay relative to V1 and reduced completion. The same direction holds in the informative d10 and k10 sweeps. At d4 and k1, heavy censoring makes completed-only distribution metrics and apparent RMST equivalence non-diagnostic. Across the five tested cells, the literal spectral-lag and latent-proxy formulas systematically over-correct the baseline timing discrepancy.

12 What the experiment can identify

It can estimate relative bias reduction in the sparse baseline, distinguish H1 from H2 within the factorial cells, assess their interaction, and describe limited scaling across the specified d and k values.

13 What the experiment cannot identify

It cannot identify recovery effects, synchronization-interval effects, placement effects, generalization beyond connected ER communities, or computational speedup. Runtime is an implementation diagnostic only.

14 Limitations

Lag may interact with midpoint hazard integration, and latent relaxation may interact with the exponential-clock residual update. The spectral formula is a candidate scaling law over a small descriptor range. Completed-only Wasserstein statistics condition on arrival and therefore must be read beside completion and all-run RMST. They are not used to claim improvement when completion probabilities are low or materially different. V1 uses the production-aligned experimental scheduler; the formula gates establish identity among experimental scheduler variants, not direct pathwise identity with the production orchestrator. The k and d mini-sweeps are one-factor-at-a-time and do not identify an interaction or a general scaling law.

15 Unresolved questions

The current design does not determine whether any remaining discrepancy is caused by synchronization timing, placement, or non-ER boundary geometry.

16 Recommended next experiment

No recommendation is made outside the staged Experiment 05 evidence. Any follow-up specified here after validation must preserve pairing and separately vary only one previously fixed mechanism.

17 Exact reproduction commands

```
python -m two_layer_ctmc.experiments.experiment_05_spectral_corrections.cli \
  --stage smoke --scenario baseline --replications 5 --master-seed 270000
python -m two_layer_ctmc.experiments.experiment_05_spectral_corrections.cli \
  --stage validation --scenario all --replications 20 --resume --master-seed 270000
# Run only after explicit approval:
python -m two_layer_ctmc.experiments.experiment_05_spectral_corrections.cli \
  --stage final --scenario all --replications 100 --resume \
  --validation-approved --master-seed 270000
```